Domestic and home automation- Energy and water use

Energy and water supply consumption monitoring to obtain advice on how to save cost and resources. Reducing energy use in your home saves you money, increases our energy security, and reduces the pollution that is emitted from non-renewable sources of energy. If you are planning to install a small renewable energy system to make your own electricity, such as a solar electric system or small wind turbine, reducing your electricity loads is the first step because it allows you to purchase a smaller and less expensive system. There are many ways you can reduce electricity use in your home

Intrusion detection systems

An intrusion alarm system is a system whose aim is to monitor and detect unauthorized access to a building. These systems are used for different purposes and in different contexts both residential or commercial. The main purpose of an intrusion alarm system is to protect from burglary, property damage and of course the security of the individuals inside the building. There are many different types of intrusion alarm systems that can be standalone elements (like a glass-break sensor) or fully incorporated security ecosystems with CCTV cameras and access control. However, for the sake of clarity, we have dedicated separate overviews to access control and video surveillance systems.

Smart City

A smart city is a technologically modern urban area that uses different types of electronic methods, voice activation methods and sensors to collect specific data. Information gained from that data are used to manage assets, resources and services efficiently; in return, that data is used to improve the operations across the city.

This includes data collected from citizens, devices, buildings and assets that is then processed and analysed to monitor and manage traffic and transportation systems, power plants, utilities, water supply networks, waste, crime detection, information systems, schools, libraries, hospitals, and other community services

Smart Parking

Smart Parking is a parking strategy that combines technology and human innovation in an effort to use as few resources as possible—such as fuel, time and space—to achieve faster, easier and denser parking of vehicles for the majority of time they remain idle

In a nutshell, Smart Parking is a parking solution that can include in-ground Smart

These devices are usually embedded into parking spots or positioned next to them to detect whether parking bays are free or occupied. This happens through real-time data collection. The data is then transmitted to a smart parking mobile application or website, which communicates the availability to its users. Some companies also offer other in-app information, such as parking prices and locations. This gives you the possibility to explore every parking option available to you.

Structural health monitoring (SHM)

It involves the observation and analysis of a system over time using periodically sampled response measurements to monitor changes to the material and geometric properties of engineering structures such as bridges and buildings

Noise urban maps

A strategic noise map (SNM) is therefore primarily requited allowing to visualize the “decibel” impact of the main sources of environmental noise either at the scale of an agglomeration or at the scale of an transportation or industrial infrastructure.

A comprehensive noise action plan (NAP) is therefore drawn with the involvement of transportation, planning, and acoustic engineers to access and specify the most appropriate means to achieve the needed noise rehabilitation by mitigation measures.

Electromagnetic radiation

Electric and magnetic fields together are referred to as electromagnetic fields, or EMFs. The electric and magnetic forces in EMFs are caused by electromagnetic radiation.

There are two main categories of EMFs: **Higher-frequency EMFs, which include x-rays and gamma rays**. These EMFs are in the ionizing radiation part of the electromagnetic spectrum and can damage DNA or cells directly. **Low- to mid-frequency EMFs, which include static fields (electric or magnetic fields that do not vary with time**), magnetic fields from electric power lines and appliances, radio waves, microwaves, infrared radiation, and visible light. These EMFs are in the non-ionizing radiation part of the electromagnetic spectrum and are not known to damage DNA or cells directly. Low- to mid-frequency EMFs include extremely low frequency EMFs (ELF-EMFs) and radiofrequency EMFs.

Smart lighting

Smart lighting is a lighting technology designed for energy efficiency, convenience and security. This may include high efficiency fixtures and automated controls that make adjustments based on conditions such as occupancy or daylight

Smart Roads

Smart roads use Internet of Things (IoT) devices to make driving safer, more efficient, and in line with government objectives, greener.

Smart roads combine physical infrastructures such as sensors and solar panels with software infrastructure like AI and big data.

Smart road technologies are embedded in roads and can improve visibility, generate energy, communicate with autonomous and connected vehicles, monitor road conditions, and more.Here are a few examples: IoT connectivity: Cities can connect roads to IoT devices, and gather traffic and weather data. This type of connectivity can improve safety, traffic management, and energy efficiency.

Smart Environment

Smart environments have the potential to allow users to engage and interact seamlessly with their immediate surroundings. This has been made possible by the introduction of intelligent technologies, coupled with software-based services.

Forest fire detection

Forest fires (wildfires) are common hazards in forests, particularly in remote or unmanaged areas. It is possible to detect forest fires, elevated CO_2, and temperature levels using various sensors.When forest fires burn, they emit large volumes of carbon dioxide gas (CO_2); you can use CO_2 and temperature sensors for forest fire detection. CO_2 and temperature sensors can offer an inexpensive alternative to help detect forest fires.

Air pollution control

Air pollution control, the techniques employed to reduce or eliminate the emission into the atmosphere of substances that can harm the environment or human health. The control of air pollution is one of the principal areas of pollution control, along with wastewater treatment, solid-waste management, and hazardous-waste management.

Snow level Monitoring

Snow depth is most often measured using an ultrasonic or radar sensor. The sensor is placed on an arm outstretched over the area to be monitored and the signal of the sensor is bounced off the snow surface. An ultrasonic sensor is employed at a place, where the level of snow is to be measured. This sensor senses the snow at that place and measures its depth by keeping ground as a base. The level of snow is continuously measured using ultrasonic sensor

Earthquake Early Detection

An earthquake early warning (EEW) system is designed to detect an event, determine its parameters (hypocenter, magnitude, and origin time), and issue an alert to sites/areas where necessary actions should be taken before destructive seismic energy arrivals. At present, largescale EEW systems are operational in several countries around the world.

Smart water- Potable water monitoring

With the rapid development of industry and agriculture, water pollution is found everywhere, and the protection of water resources has attracted increasing attention. For a long time, drinking water pollution was measured manually, which is time-consuming and laborious. To effectively detect and evaluate drinking water pollution, a drinking water quality monitoring and evaluation system is designed. **The system can perform real-time measurements of water temperature, conductivity, turbidity and other parameters.**

Chemical leakage detection in rivers

The water that runs through the environment carries the stories of the land. Each drop contributes to an ecosystem both vast and complex and to a watershed we all depend on. Our reliance on this system connects us — the plants, the frogs, the fish and the people. But it's easy to forget how closely we are connected, easy to ignore the way the water is affecting all life forms, and how much our own well-being depends on it. Extreme pH or low DO values signal chemical spills due to sewage treatment plant or supply pipe problems. The Smart Water wireless sensor platform is simplifying remote water quality monitoring.

Equipped with multiple sensors that measure a dozen of the most relevant water quality parameters.

Swimming pool remote measurement

Measuring oxidation-reduction potential (ORP), pH and Chloride levels of water can determine if the water quality in swimming pools is sufficient for recreational purposes.

The Smart Water platform is an ultra-low-power sensor node designed for use in rugged environments and deployment in Smart Cities in hard-to-access locations to detect changes and potential risk to public health in real time.

It may use cellular (3G, GPRS, WCDMA) and long range 802.15.4/ZigBee (868/900MHz)connectivity to send information to the Cloud, and can accommodate solar panels that charge the battery to maintain autonomy.

SMART METER

A **smart meter is an electronic device that records information such as consumption of electric** energy, voltage levels, current, and power factor. Smart meters communicate the information to the consumer for greater clarity of consumption behaviour, and electricity suppliers for system monitoring and customer billing. Smart meters typically record energy near real-time, and report regularly, short intervals throughout the day.

Tank Level Monitoring System

Sensors used in tank level monitoring are used to identify the levels of tank for a variety of liquids. It allows different users to track the tank levels remotely without any physical human efforts like manual meter reading and assessments. Tank level monitoring provides a greater insight for tank depletion rates and also helps in increasing the efficiency of inventory management. Manual inspection of tank level are time consuming, error prone, expensive, unsafe, inaccurate. So, there is a high demand for remotely accessing of the tank levels for monitoring the levels of tank in less time.

Liquid presence Detection system

Leaks in pipelines and Other Systems that involve water, gas or chemicals present a serious issue With the potential for significant economic losses or substantial environmental pollution.

Liquid presence detection Systems are commonly used to monitor leaks. but in many cases, they are placed at long intervals along a pipeline due to the high Cost Of deployment.

NFC payment, smart product management

Contactless payment and multi-application solutions for cards, smart accessories and wearables and other upcoming new form factors. The move away from cash and contact-based payment cards continues to accelerate. Contactless "tap and go" transactions using cards, smart wearables or mobile devices are increasingly replacing cash and contact-based transactions.

Many experts have earmarked payment as the "killer app" for wearables. Customers are looking for flexible enablement platforms to support multi-application payment cards, supporting domestic payment schemes and project-specific requirements with the possibility to integrate multi-applications functionalities like ticketing applications.